Jordan Dialpuri

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Postdoctoral research scientist at the MRC Laboratory of Molecular Biology, Cambridge and co-founder of data visualisation company with a PhD in Biological Chemistry. Research focuses on development of novel deep learning based methods for structural biology with integrations to the popular software package Coot. Expertise in engineering predictive models for three-dimensional biological data, with a passion for scientific communication and web technologies.

Experience

Postdoctoral Research Scientist April 2026 – Present
MRC Laboratory of Molecular Biology *Cambridge*

- Integrating deep learning methods developed during doctorate into Coot to advance model-building capabilities for nucleic acid and carbohydrates.
- Combined quantum chemical calculations of conformational states with Bayesian statistics to aid automated model building of carbohydrates.
- Implemented command line interface to Coot to increase efficiency in structural modelling and visualisation.
- Invited to teach Coot and model building at international structural biology workshop.

Co-founder January 2024 – Present
Nerate Ltd *York*

- Co-founded data visualisation training company offering in-person and online training to researchers in academia.
- Provided training to over 10 academic institutions in the United Kingdom and United States.
- Engineered all website and cloud infrastructure for custom online training platform.

Placement Student June 2021 – September 2022
Afton Chemical *Bracknell*

- Placement as part of fourth year of undergraduate degree, undertaking a research project in engine oil additive optimisation. Implementing novel machine learning models to predict the outcome of £150,000 industrial test.

Education

Doctor of Philosophy in Biological Chemistry
University of York *October 2022 – April 2026*

- Title: Automated Model Building of Nucleic Acids and Carbohydrates Using Experimental Data and Deep Learning Models.
- Produced deep learning models which simulate the intuition of an expert crystallographer at understanding experimental data.
- Developed efficient automated software packages to produce accurate, high-quality macromolecular structures of nucleic acids and carbohydrates.
- Created innovative web application to allow easy and intuitive online carbohydrate structure validation.
- Presented machine learning work at many national and international conferences.
- Invited to teach structural biology at multiple structural biology workshops.

MChem in Chemistry **First Class Honours with Distinction (1*)**
University of York *September 2018 - July 2022*

Select Publication

***NucleoFind*: a deep-learning network for interpreting nucleic acid electron density** August 2024

- Produced three-dimensional U-Net model using TensorFlow to predict the positions of the phosphate group, sugar ring and nitrogenous base group in three-dimensional electron density.
- Transformed predicted positions into a three-dimensional model of nucleic acid automatically, reducing the need for manual modelling.
- Software produces more complete and accurate molecular models than existing methods, shortening nucleic acid structure solution from weeks or months to hours.
- Open source software written in Python and C++ and distributed on PyPi.
- Citation: **J. S. Dialpuri**, J. Agirre, K. D. Cowtan and P. S. Bond, Nucleic Acids Research, 2024, 52, e84

Technical Skills

Languages: Python, C++, JavaScript, R, Swift
Libraries: TensorFlow, PyTorch, React
Courses: Nvidia GPU programming

Awards

Margaret C. Etter Student Lecturer Award for exceptional scientific quality and potential in the field of crystallography

Kathleen Mary Stott Prize for excellence in scientific research